

# When Blocking Earns Its Keep

Unit 1 · Topic 1.13 · TODAY'S PROBLEM

Suppose a grower has 24 similar tomato plants, 12 in a sunny greenhouse and 12 in a shaded greenhouse. The response is fruit yield in kilograms per plant.

**Design A:** Completely randomize all 24 plants; assign 12 to fertilizer and 12 to no fertilizer, ignoring greenhouse.

**Design B:** Treat greenhouse as a block; within each greenhouse, randomly assign 6 plants to fertilizer and 6 to no fertilizer.

Pilot means from plants grown under the same conditions are shown.

Mean fruit yield (kg per plant)

| Greenhouse | Fertilizer | None |
|------------|------------|------|
| Sunny      | 7.2        | 6.0  |
| Shaded     | 4.0        | 2.8  |

## FIRST TAKE (5 min, on your sheet)

Which design would you use to detect the fertilizer effect most clearly? Name one detail behind your choice.

- DOK 1** (a) Identify the treatments, response variable with units, and blocking variable in Design B.
- DOK 2** (b) Compute the fertilizer-minus-control difference within each greenhouse and the sunny-minus-shaded difference within each treatment.
- DOK 3** (c) Which design should the grower use? Cite the relative sizes of the greenhouse and fertilizer differences that settle your choice, and explain what chance imbalance could do to Design A. If Design B produced a statistically significant fertilizer difference, state the causal conclusion it would permit and one limit on generalizing it.

**Video + follow-along:** [u3\\_lesson6-7\\_live.html](https://u3_lesson6-7_live.html) (scan the code)

Finish (a)–(c) after the video. Turn in your sheet.

